

**A  
MINI PROJECT**

**On**

**PEER TO PEER ENGAGEMENT ATTRIBUTION AND COLLABORATIVE  
LEARNING ANALYTICS MODEL**

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## ABSTRACT

Traditional skill certification systems rely on centralized authorities and static documents, which are prone to forgery and lack real-time verification. This project proposes a decentralized skill credentialing platform where verified skill records are stored in a tamper-resistant ledger. Each skill credential issued to a student is cryptographically recorded and can be independently verified without relying on a single authority. The system ensures transparency, trust, and long-term integrity of skill credentials while enabling institutions and recruiters to validate authenticity instantly through a secure verification gateway.

The proposed model integrates data from multiple interaction sources such as discussions, shared tasks, peer feedback, and contribution frequency. Using analytical techniques and scoring algorithms, the system assigns weighted engagement scores to each participant, ensuring fair recognition of individual efforts. Additionally, it provides visual insights and performance reports that help educators monitor group dynamics and identify active and passive learners.

By promoting transparency and accountability, this model encourages meaningful participation and improves learning outcomes. It also supports personalized feedback and data-driven decision-making in academic environments. Overall, the system aims to transform collaborative learning into a more measurable, fair, and effective educational experience.

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# **1 INTRODUCTION**

## **1.1 Introduction to the project**

In modern online and blended learning environments, students frequently interact with peers through discussions, group work, and collaborative activities. However, most existing learning systems mainly focus on individual performance and teacher-led assessments, while ignoring peer-to-peer engagement and its impact on learning outcomes. This project proposes a Peer-to-Peer Engagement Attribution and Collaborative Learning Analytics Model that tracks, analyzes, and attributes student participation during collaborative learning activities. The system collects interaction data such as messages, replies, shared resources, and feedback among learners. Using learning analytics techniques, it measures engagement levels, contribution quality, and collaboration patterns.

The proposed model helps educators understand how students learn together, identify active and passive learners, and promote effective collaboration. It also provides students with feedback on their participation, encouraging equal contribution and teamwork. The system is easy to implement and suitable for online classrooms, discussion forums, and virtual group projects.

## **1.2 Background**

Peer-to-Peer Engagement Attribution is a concept used in collaborative learning systems and online education platforms to understand how students influence each other's learning. In traditional classrooms, learning mainly happens between the teacher and the student. However, in modern digital learning environments, students also learn a lot from interactions with their peers through discussions, group activities, forums, and shared resources.

The idea of peer-to-peer engagement attribution focuses on tracking and measuring how much one student's participation helps another student learn or stay engaged. For example, when a student asks a question in a discussion forum and another student answers it, the system can record that interaction. If the answer helps the first student understand the concept or complete an assignment, the system attributes part of that learning outcome to the helping student.

### **1.3 Objective of the project**

To track peer-to-peer interactions in collaborative learning environments.

To measure student engagement levels during group activities.

To fairly attribute individual contributions in team-based learning.

To analyze collaborative learning behavior using data analytics.

To identify active and passive learners.

To provide meaningful insights to teachers for better evaluation.

### **1.4 Scope of the project**

The scope of the Peer-to-Peer Engagement Attribution project focuses on analyzing and measuring how students interact and support each other in a collaborative learning environment. The project aims to develop a system that tracks student activities such as discussions, comments, resource sharing, and collaborative tasks in online learning platforms.

One major scope of this project is to collect and analyze engagement data from students. The system records interactions between students, such as answering questions, participating in group discussions, or providing feedback. This data helps in identifying how peer interactions influence learning outcomes and student participation.

## **1.5 Organization of the Report**

The report on Peer-to-Peer Engagement Attribution is organized into several sections to provide a clear understanding of the project. The first section is the Introduction, which explains the background, purpose, and importance of peer-to-peer engagement in collaborative learning environments. The next section is the Literature Review, which discusses previous studies and research related to peer learning, student engagement, and learning analytics. After that, the Methodology section describes the approach used in the project, including data collection methods, tools, and techniques used to analyze peer interactions. The report also includes a System Design or Implementation section that explains how the engagement attribution system works and how it tracks student interactions.

This system integrates advanced analytics with user-friendly interfaces to monitor participation, evaluate individual and group performance, and provide meaningful insights into learning behaviors. By leveraging data such as discussion frequency, contribution quality, peer feedback, and collaboration patterns, the model ensures fair recognition of each student's efforts while promoting active engagement and accountability. It supports educators in identifying high-performing contributors, inactive participants, and overall group dynamics, enabling informed decision-making and personalized intervention strategies. Additionally, the system encourages transparency and motivation among learners by providing real-time feedback and performance dashboards. Built using modern full-stack technologies, the model is scalable, efficient, and adaptable to various educational settings, making it a valuable tool for improving collaborative learning outcomes and fostering a more interactive, data-driven academic ecosystem.

## **2 LITERATURE SURVEY**

### **2.1 Existing system**

Existing models for peer-to-peer (P2P) engagement in collaborative learning primarily focus on qualitative analysis of interactions and the use of general learning analytics (LA) to identify interaction patterns rather than explicit, universally apted attribution models for individual learning outcomes.

### **2.2 Disadvantages of Existing System**

Skills are mentioned only in resumes

No verification of skill authenticity

Fake or exaggerated skills possible

Recruiters rely on interviews only

No centralized skill tracking

No faculty involvement in validation

Skill growth is not measurable

Students don't know their real strength level

Portfolios lack institutional approval

No credibility scoring mechanism

## **2.3 Proposed System**

The proposed systems and future research directions focus on developing more robust, context- sensitive, and automated methods for collaboration analytics, often incorporating AI and multimodal data.

The proposed system collects data from different peer activities such as posting questions, replying to discussions, sharing learning materials, and participating in collaborative tasks. This data is then processed to measure the level of engagement and identify the contribution of each student in helping others learn.

## **2.4 Advantages of Proposed System**

Students upload skill-wise proof

Proof types include GitHub, images, PDFs

Faculty verifies or rejects proofs

Each skill has approval status

Rule-based credibility score generation

Skill progress history tracking

Verified skill badges provided

Public shareable profile link

Cloud-based access from anywhere

Reduces fake skill claims

## **3 SOFTWARE REQUIREMENTS ANALYSIS**

### **3.1 Problem Definition**

In modern online and collaborative learning environments, students frequently interact with each other through discussion forums, group activities, and peer feedback systems. These interactions play an important role in improving learning outcomes and encouraging knowledge sharing. However, most traditional learning management systems mainly focus on teacher–student interaction and do not effectively measure or recognize the impact of student-to-student engagement.

The main problem is the lack of a systematic method to track, analyze, and attribute the influence of peer interactions on student learning and participation. It is difficult for educators to identify which students are actively helping others, how peer discussions contribute to understanding concepts, and how collaborative learning affects overall engagement.

### **3.2 Functional Requirements**

#### **User Registration and Login**

The system should allow students and educators to create accounts and securely log in to access the platform.

#### **User Profile Management**

Users should be able to create and manage their profiles, including basic details and learning preferences.

#### **Discussion and Interaction Module**

The system should allow students to participate in discussion forums, ask questions, answer queries, and share knowledge with peers.

#### **Activity Tracking**

The system should track all peer interactions such as comments, replies, likes, and shared

resources.

### **Engagement Data Collection**

The platform should collect and store data related to student participation and peer-to-peer communication.

## **3.3 Non-Functional Requirements**

### **Performance**

The system should process and analyze peer interaction data quickly and provide engagement results or dashboard updates with minimal delay.

### **Scalability**

The system should be able to handle a large number of users, interactions, and data as the number of students and activities increases.

### **Reliability**

The system must operate consistently without failures and ensure accurate tracking of peer interactions and engagement data.

### **Security**

The system should protect student data and ensure that only authorized users (teachers or administrators) can access sensitive engagement information.

### **Usability**

The interface should be simple and user-friendly so that both students and educators can easily use the system and understand the engagement analytics.

## **3.4 System Requirements**

### **User Authentication System**

The system should allow users (students and educators) to securely register and log in to access the platform.

### **User Profile Management**

Each student should have a profile where their participation, activities, and engagement

records are stored.

#### **Discussion and Interaction Module**

The system should support peer-to-peer interactions such as discussion forums, comments, replies, and question–answer activities.

#### **Data Collection Module**

The system must collect data related to student activities such as posts, replies, likes, shares, and collaborative tasks.

#### **Engagement Tracking System**

It should track and record how frequently students interact with each other and measure their level of participation.

### **3.5 Modules Descriptions**

#### **Authentication Module**

The Authentication Module is an essential component of the system that ensures secure access for users. It manages the processes of user registration, login, identity verification, and access control within the application. This module allows users such as students and educators to create accounts, securely log into the system, and access features based on their assigned roles. The main objective of the authentication module is to protect user data, prevent unauthorized access, and maintain system security...

#### **Contribution Tracking Module**

The Contribution Tracking Module is responsible for monitoring, recording, and analyzing the activities performed by users within the system. This module plays an important role in evaluating how actively users participate and collaborate on the platform. It collects information about different types of user contributions such as discussions, peer reviews, code commits, and resource sharing. By tracking these activities, the system can measure the level of engagement of each user and use this data for calculating contribution scores in the overall engagement scoring algorithm.

#### **Scoring Engine Module**

The Scoring Engine Module is an important component of the system that is responsible for evaluating and calculating the level of user participation and engagement within the platform. This module analyzes different user activities and assigns scores based on predefined rules. The main objective of the scoring engine is to measure user involvement, encourage active participation, and provide a fair evaluation of user contributions. By using a structured scoring system, the platform can identify highly active users and motivate others to participate more effectively.

### **Analytics & Ranking Module**

The Analytics & Ranking Module is responsible for analyzing user activity data and presenting meaningful insights about student performance and engagement. This module collects data generated from different system components, such as the scoring engine, user activity logs, and participation records. By processing this information, the module helps administrators, educators, and students understand performance trends and identify areas for improvement. The main goal of this module is to provide clear performance analysis, ranking systems, and detailed assessment reports that help improve learning outcomes and encourage healthy competition among students.

### **Dashboard Module**

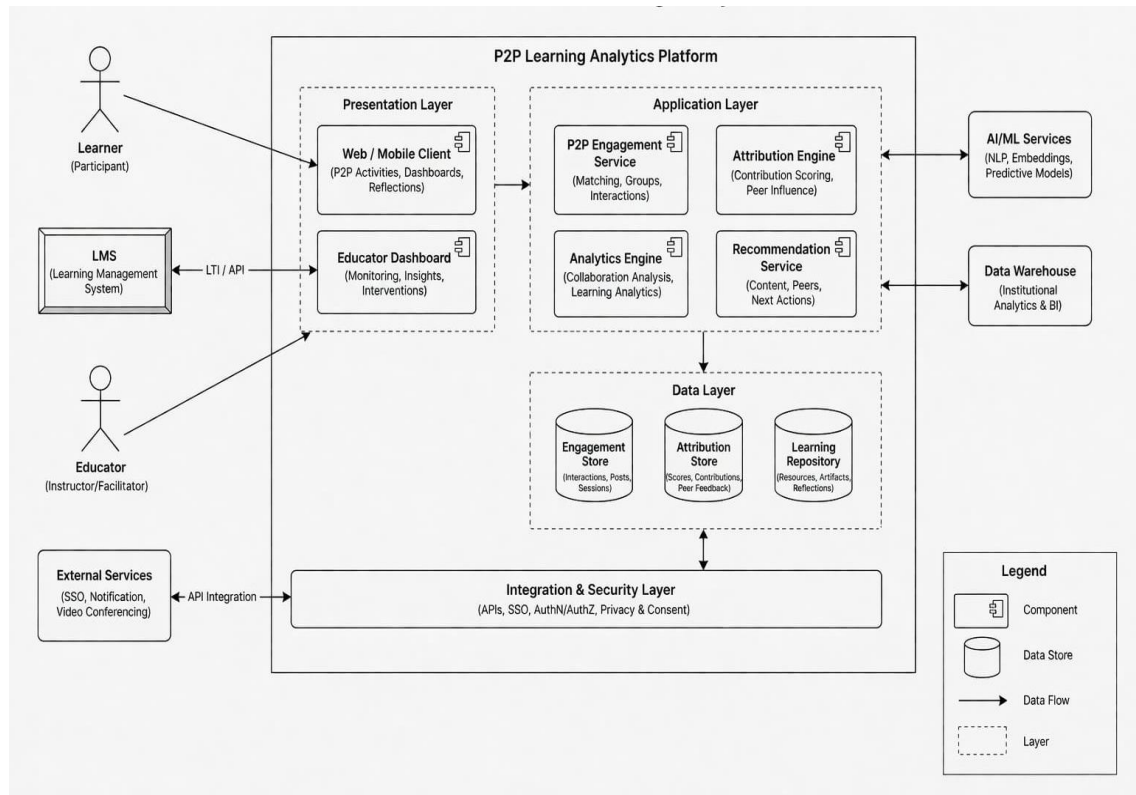
The Dashboard Module is an important part of the system that provides the modules users with a visual interface to monitor activities, performance, and engagement levels within the platform. It presents important information in a clear and organized manner using graphs, charts, and statistical summaries.

## 4 SYSTEM DESIGN

### 4.1 System Architecture

The System Architecture of the Peer-to-Peer Engagement Attribution project defines how different components of the system work together to track, analyze, and display peer interactions in a collaborative learning environment. The architecture is designed to collect engagement data from students, process it, and present meaningful insights to educators through dashboards and analytics tools.

The system mainly consists of three layers: the presentation layer, application layer, and data layer. The presentation layer is the user interface where students and educators interact with the system.

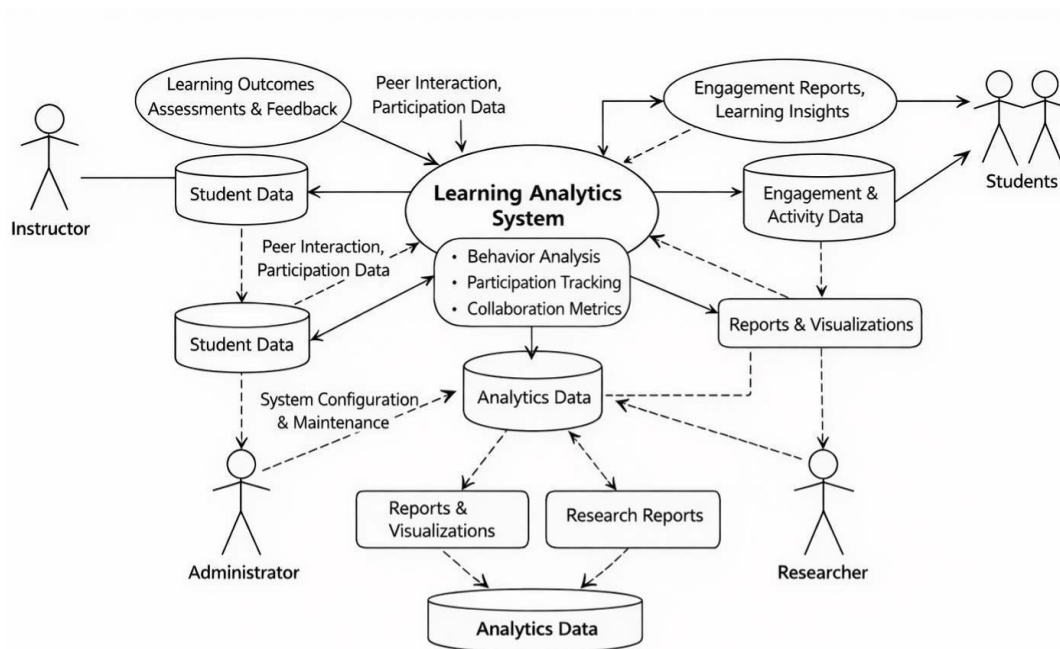


**Fig 4.1** System Architecture of peer-to-peer engagement attribution and collaborative learning analytics model.

## 4.2 Data Flow Diagram

Data Flow Diagrams are used to represent how data moves through the system. They help in understanding input, processing, storage, and output of data related to peer engagement and analytics.

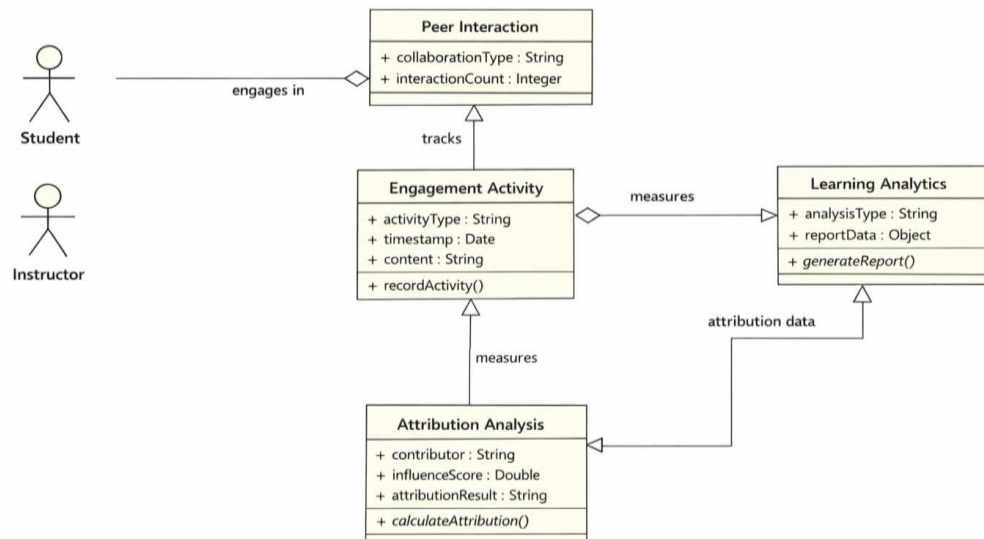
Represents the system as a single process interacting with external entities such as Student, Instructor, and Admin. Students provide peer interaction data to the system, instructors request analytics reports, and admins manage system information. The system processes these inputs and produces engagement analytics and reports.



**Fig 4.2** Data Flow diagram of peer engagement attribution and collaborative learning analytics model

### 4.3 UML Diagram

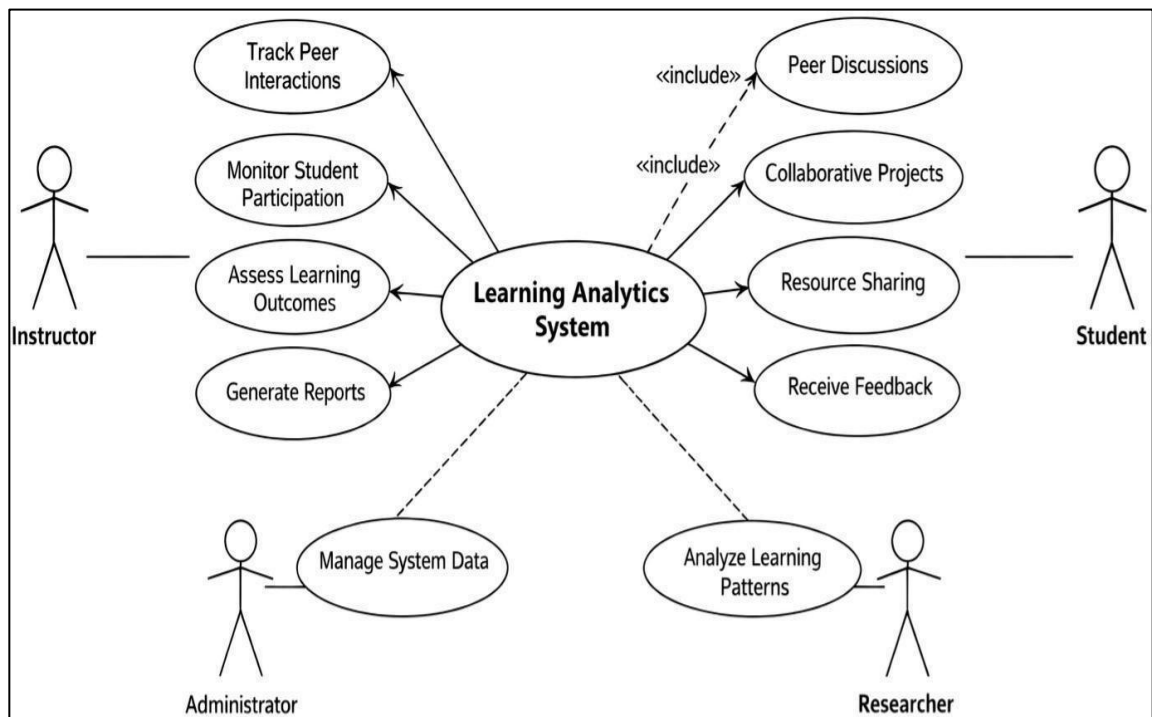
Unified Modeling Language (UML) is used to visually describe the system architecture and interactions among different components. UML diagrams provide a clear representation of system functionality, data organization, and communication between users and system modules. In this project, UML diagrams such as Use Case Diagram and Class Diagram are used to describe system behavior and structure in a standardized manner. The UML diagrams play a vital role in explaining the design and working of the Peer-to- Peer Engagement Attribution and Collaborative Learning Analytics Model. These diagrams provide a clear and structured visualization of how users interact with the system, how data flows internally, and how engagement is measured and analyzed. By using UML, the project design becomes easier to understand, implement, and maintain, especially in complex collaborative learning environments where multiple users and interactions are involved.



**Fig 4.3** UML diagram of peer-to-peer engagement attribution and collaborative learning analytics model.

## Use Case Diagram

The Use Case Diagram illustrates the interaction between external users and the system. It identifies different actors involved and the functionalities provided by the system. The main actors in this system are Student, Instructor, and Admin. Students participate in collaborative learning activities such as discussions, peer feedback, and resource sharing. Instructors monitor student engagement and analyze collaborative learning outcomes through reports. The admin manages user accounts, system configurations, and overall maintenance.



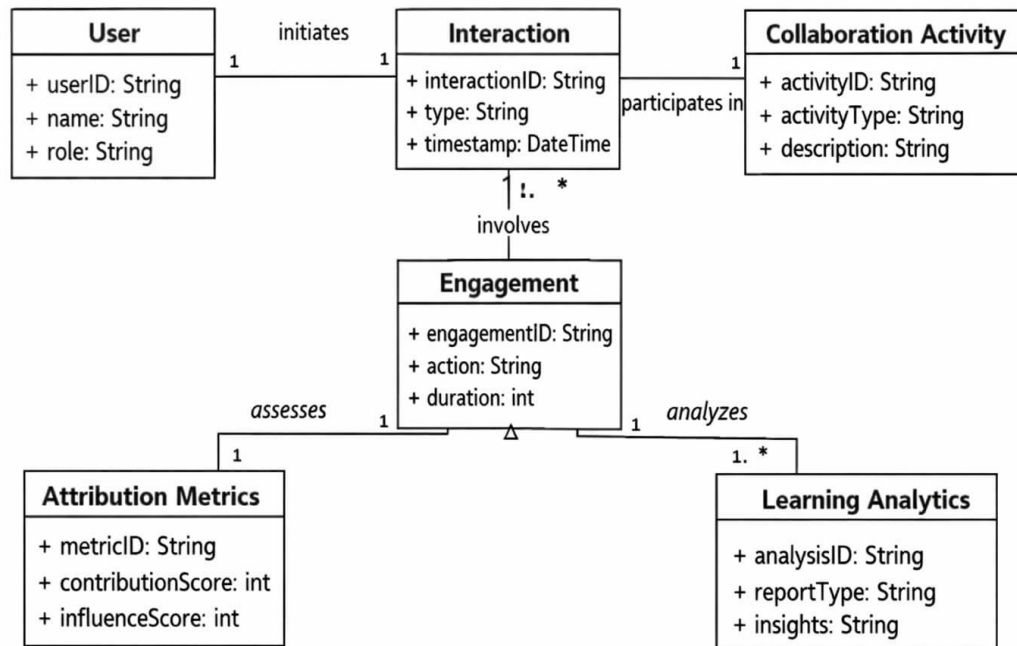
**Fig 4.3.1** USECASE diagram of peer-to-peer engagement attribution and collaborative learning analytics model.

## Class Diagram

The Class Diagram represents the static structure of the system by defining classes,

attributes, methods, and relationships among them. It explains how system data is organized and managed internally. The main classes include User, Student, Instructor, PeerGroup, PeerInteraction, EngagementAttribution, AnalyticsEngine, and Report. The User class acts as a base class, while Student and Instructor inherit its properties. The PeerInteraction class stores details of peer activities such as discussions and feedback. EngagementAttribution calculates engagement scores based on interaction data. The AnalyticsEngine processes this data and generates insights, which are presented through the Report class.

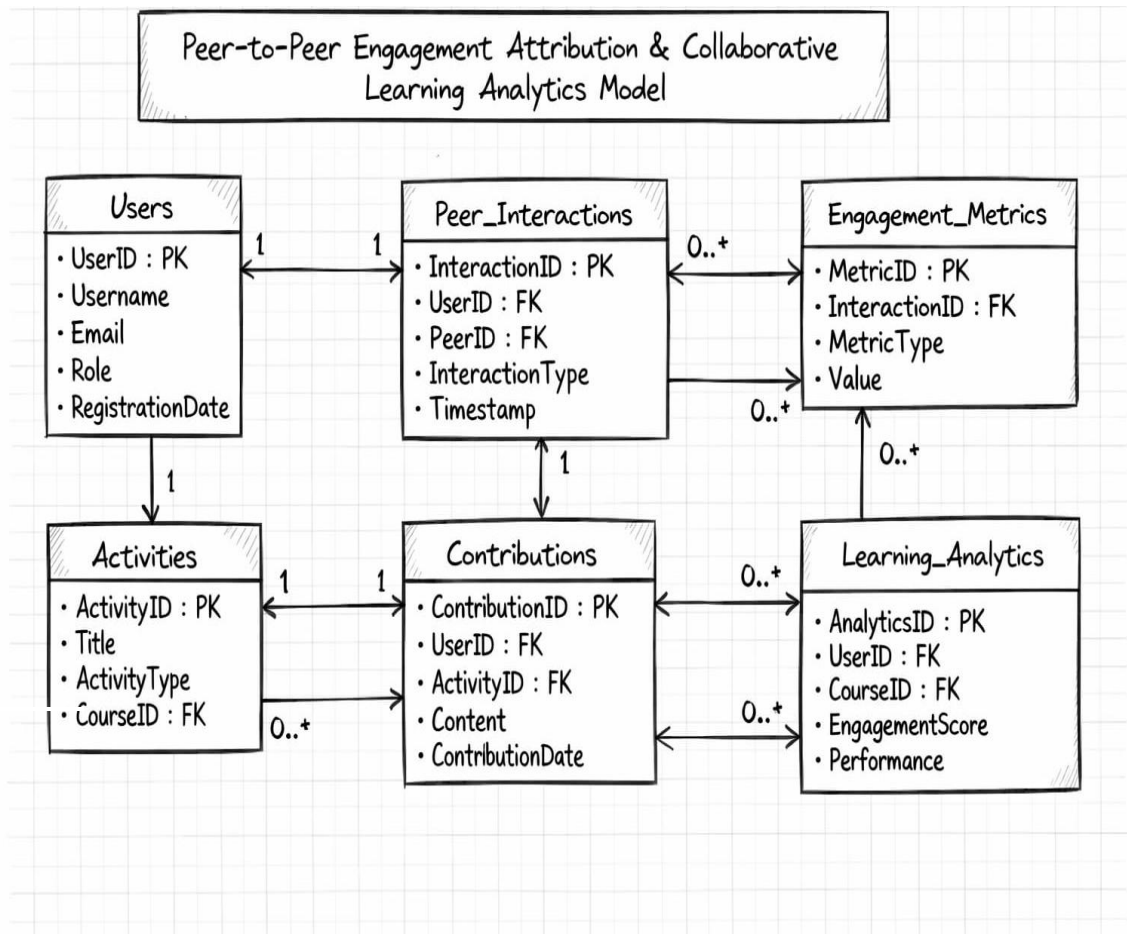
**Peer-to-Peer Engagement Attribution and Collaborative Learning Analytics Model**



**Fig 4.3.2** Class diagram of peer engagement attribution and collaborative Learning analytics model

## 4.4 Data Base Design

The database design for the Peer-to-Peer Engagement Attribution system is responsible for storing and managing all the data related to students, their interactions, and engagement activities. A well-structured database helps the system efficiently track peer-to-peer communication, analyze participation levels, and generate meaningful reports for educators.

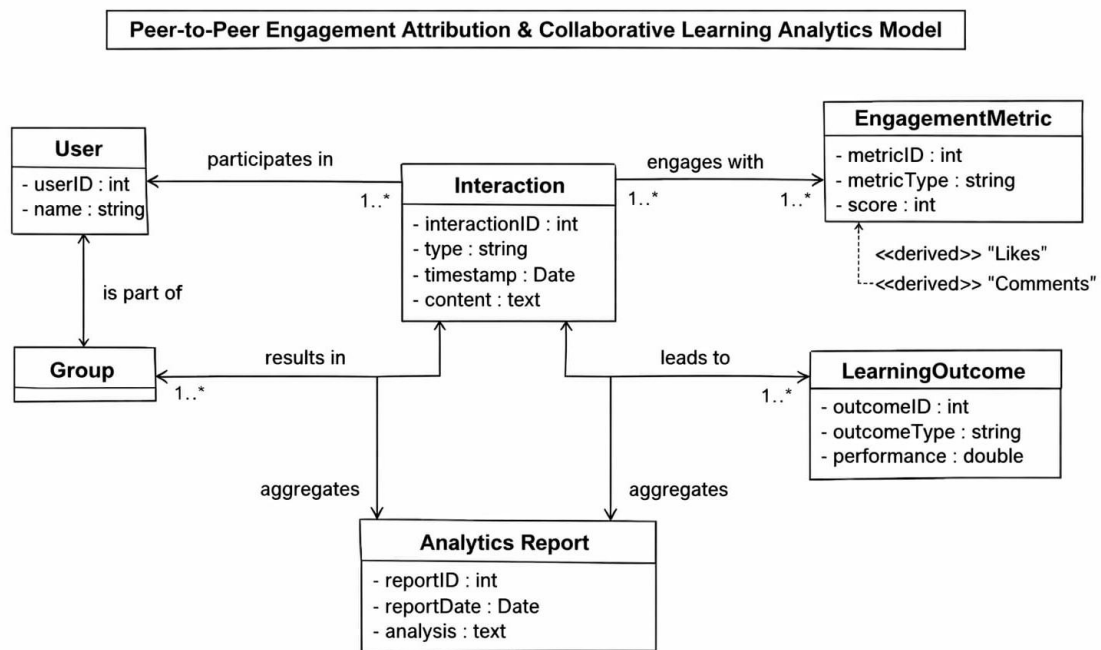


**Fig 4.4** Data Base Design of peer engagement attribution and collaborative Learning analytics model.

## 4.5 ER Diagram

The Entity Relationship (ER) Diagram in the Peer-to-Peer Engagement Attribution system represents the structure of the database and the relationships between different entities involved in the system. It helps in understanding how data related to students, interactions, and engagement activities are organized and connected within the system.

The main entity in the ER diagram is the User entity, which stores information about students and educators such as user ID, name, email, and role. Each user can participate in multiple discussions and engagement activities within the platform.



**Fig 4.5** E-R Diagram of peer-to-peer engagement attribution and collaborative analytics model.

## **5 SOFTWARE AND HARDWARE REQUIREMENTS**

### **5.1 Hardware Requirements:**

**Processor:** Intel Core i3 or above

**RAM:** Minimum 8 GB

**Hard Disk:** Minimum 20 GB free space

**System:** Desktop or Laptop

**Input Devices:** Keyboard, Mouse

**Output Device:** Monitor

### **5.2 Software Requirements:**

**Operating System:** Windows 10 / Windows 11 / Linux

**Programming Language:** Python

**Frontend:** React js, HTML, CSS

**Backend:** Python (Flask or Fast API)

**Database:** MongoDB Atlas(cloud)

**IDE / Editor:** VS Code / PyCharm / IDLE

**Web Browser:** Google Chrome / Firefox

## **6 IMPLEMENTATION/CODING**

### **6.1 Technologies Used**

#### **6.1.1 Programming Language**

Python

The entire system is developed using Python.

Python is used for authentication, analytics, database operations, and dashboard management.

#### **Database**

MongoDB

Used to store:

User information

Contributions

Collaborations

Peer ratings

The code connects to MongoDB using Py Mongo.

#### **6.1.2 Libraries / Packages Used**

##### **Py Mongo**

Used to connect Python with MongoDB database.

Helps store and retrieve engagement data.

bcrypt

Used for password hashing.

Ensures secure user authentication.

##### **Tabulate**

Used to display data in table format in the terminal.

Example: Showing contributions, ratings, or analytics.

### **python-dateutil**

Helps handle date and time operations.

## **6.1.3 Modules Used in the Project**

The system is divided into multiple modules:

### **database.py**

Handles database connection.

### **auth.py**

Manages user registration and login.

### **models.py**

Defines data models like:

Contribution

Collaboration

PeerRating

User

### **scoring\_engine.py**

Calculates engagement scores for students.

### **analytics.py**

Generates analytics and performance reports.

### **dashboard.py**

Displays dashboards for students and educators.

## 6.2 SAMPLE CODE

```
Import sys import os
# Import all modules try:
From database import get_db from auth import AuthManager
From models import Contribution, Collaboration, PeerRating, User from
scoring_engine import ScoringEngine
From analytics import Analytics from dashboard import Dashboard
From config import CONTRIBUTION_TYPES from tabulate import tabulate
Except ImportError as e: print (f' \nX Import Error: {e}')
Print (\n Please install required packages)
Print ("pip install pymongo bcrypt tabulate python-dateutil") sys. exit (1)

Class PeerEngagementSystem:
    Def init (self):
        # Initialize database try:
        Import database database.db_instance = get_db () self.db = database.db_instance
        except Exception as e:
            Print (f' \nX Failed to initialize database: {e}')
            Print (\nPlease ensure MongoDB is running on localhost:27017) sys. exit
            (1) # Initialize managers
            Self.auth = AuthManager(self.db) self. scoring_engine = ScoringEngine(self.db)
            Self.analytics = Analytics (self.db, self. scoring_engine) self. dashboard =
            Dashboard (self. auth, self. analytics) self. running = True
            Def display_banner(self):
```

```

Print (\n + "=" *70)
Print ("PEER-TO-PEER ENGAGEMENT ATTRIBUTION SYSTEM")
Print ("Collaborative Learning Analytics Model") print ("=" *70)
Def main_menu(self):
Print (\n + "-" *70) print ("MAIN MENU") Print
("-" *70) print (\n1. Register) print ("2. Login")
Print ("3. Exit")
Choice = input ("\nSelect option: "). strip () return choice
Def authenticated_menu(self):
Current_user = self.auth.get_current_user () if not
current_user: Print (\nSession expired. Please login again.)
Self.auth. logout ()
return Print (f\n
{'-'*70})
Print (f" Welcome, {current_user['username']} ({current_user['role']. upper ()})") print (f"
{'-'*70}")
If current_user['role'] == 'student': self. student_menu ()
Elif current_user['role'] == 'educator': self. educator_menu ()
Def student_menu(self):
while True:
Print (\n1. View My
Dashboard) Print ("2. Log
Contribution") print ("3. Log
Collaboration") print ("4. Rate
a Peer")
Print ("5. View My Contributions") print
("6. View My Collaborations") print ("7.
Logout")

```

```

Def
register_user(self):
    Print (\n + “—” *70)
    Print (“USER REGISTRATION”)
    Print (“—” *70)
    Username = input (“\nEnter username: “). strip ()
    email = input (“Enter email: “). strip ()
    Password = input (“Enter password: “). strip ()
    Print (\nSelect role:)
    print (“1. Student”)
    print (“2. Educator”)
    Role_choice = input (“Enter choice (1/2): “). strip
    () If role_choice not in [‘1’, ‘2’]:
    Print (\nInvalid role selection!)
    return
    Role = ‘student’ if role_choice == ‘1’
    else
    ‘educator’ department =
    ‘’ If role == ‘student’:
    Department = input (“Enter department: “). strip ()
    Success, message = self. auth. register (username, email, password, role, department)
    print(f\n{message})

```

## **7 TESTING**

Testing is an essential phase in the development of the Peer-to-Peer Engagement Attribution and Collaborative Learning Analytics Model. It ensures that the system works correctly, performs efficiently, and meets the required objectives of tracking student interactions and analyzing collaborative learning behavior

### **7.1 Testing Objectives**

The main objective of testing the Peer-to-Peer Engagement Attribution and Collaborative Learning Analytics Model is to ensure that the system works correctly, processes data accurately, and provides reliable analytics about student collaboration and dr accurate and fairly represented.

### **7.2 Test Cases**

#### **Unit Testing**

Unit testing is the process of testing individual components or modules of the system separately to verify that each unit performs its intended function correctly. In this project, different modules such as the authentication module, scoring engine module, analytics module, and dashboard module are tested individually.

#### **Integration Testing**

Integration testing focuses on verifying the interaction between different modules of the system. After individual modules are tested through unit testing, they are combined and tested together to ensure that they communicate properly and exchange data accurately.

## System Testing

System testing is performed to evaluate the complete and fully integrated system. In this stage, the entire application is tested as a whole to verify that all features function according to the project requirements.

## Performance Testing

Performance testing is conducted to evaluate how well the system performs under different workloads and user activities. Since the system tracks peer interactions and calculates analytics in real time, it is important to ensure that the system can handle multiple users and large amounts of data without slowing down.

## Acceptance Testing

Acceptance testing is the final stage of the testing process, where the system is evaluated by end users or project supervisors to ensure that it meets the intended requirements. In the Peer-to-Peer Engagement Attribution and Collaborative Learning Analytics.

### 7.2.1 Login Module Testing

| Test Case ID | Module | Test Scenario        | Input                         | Output           | Status |
|--------------|--------|----------------------|-------------------------------|------------------|--------|
| TC_01        | Login  | Valid student login, | Correct username and password | Login successful | Pass   |

|       |       |                    |                                     |                  |      |
|-------|-------|--------------------|-------------------------------------|------------------|------|
| TC_02 | Login | Invalid password   | Correct username and wrong password | Error displayed  | Pass |
| TC_03 | Login | Empty login fields | Blank username and password         | Validation shown | Pass |
| TC_04 | Login | Admin login        | Valid admin credentials             | Dashboard opened | Pass |

### 7.2.2 Registration Module Testing

| Test Case ID | Module       | Test Scenario                | Input                 | Output             | Status |
|--------------|--------------|------------------------------|-----------------------|--------------------|--------|
| TC_05        | Registration | New student registration     | Valid student details | Account created    | Pass   |
| TC_06        | Registration | Duplicate email registration | Existing email ID     | Error displayed    | Pass   |
| TC_07        | Registration | Missing mandatory fields     | Blank required fields | Validation shown   | Pass   |
| TC_08        | Registration | Invalid mobile number        | Less than 10 digits   | Failed as expected | Pass   |

### 7.2.3 Analytics Module Testing

| Test Case ID | Module    | Test Scenario                 | Input            | Output           | Status |
|--------------|-----------|-------------------------------|------------------|------------------|--------|
| TC_09        | Analytics | View engagement score         | Valid student ID | Score displayed  | Pass   |
| TC_10        | Analytics | Generate collaborative report | Valid group data | Report generated | Pass   |
| TC_11        | Analytics | Invalid student ID            | Wrong student ID | Error displayed  | Pass   |

### 7.2.4 Logout Module Testing

| Test Case ID | Module | Test Scenario          | Input                       | Output            | Status |
|--------------|--------|------------------------|-----------------------------|-------------------|--------|
| TC_12        | Logout | Student logout         | Click logout button         | Logout successful | Pass   |
| TC_13        | Logout | Session timeout logout | User inactive for long time | Auto logout done  | Pass   |

### 7.3 Black Box Testing

Black Box Testing is a software testing method where the tester checks the functionality of the system without knowing the internal code or program structure. The tester only focuses on inputs and expected outputs of the system.

| Test Case ID | Module Name       | Input Provided                      | Expected Output                                       | Actual Output                      | Status |
|--------------|-------------------|-------------------------------------|-------------------------------------------------------|------------------------------------|--------|
| TC01         | User Registration | Valid username, email, and password | User account should be created successfully           | User account created successfully  | Pass   |
| TC02         | User Registration | Invalid email format                | System should display error message for invalid email | Error message displayed            | Pass   |
| TC03         | User Login        | Correct username and password       | User should log in successfully                       | Dashboard opened successfully      | Pass   |
| TC04         | User Login        | Incorrect password                  | System displays invalid password                      | Invalid password message displayed | Pass   |

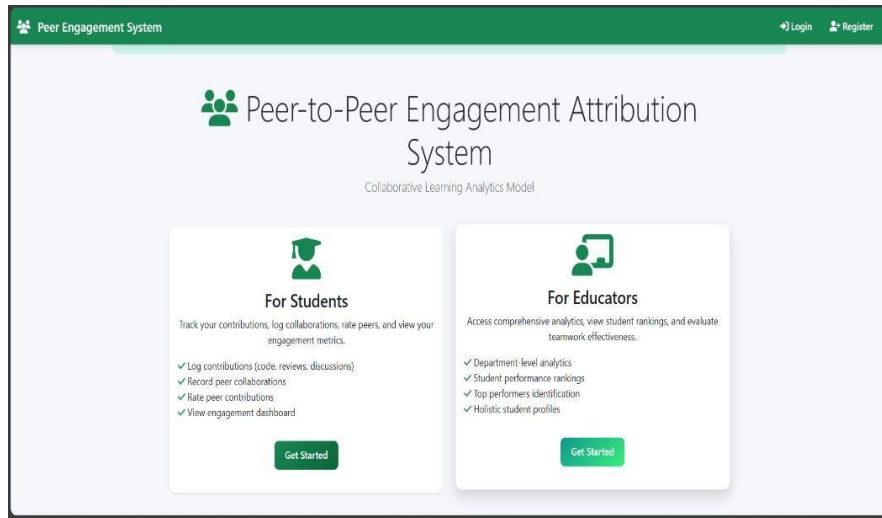
## 7.4 White Box Testing

White Box Testing is a software testing technique in which the internal structure, logic, and code of the system are tested. The tester has knowledge of the source code, algorithms, and program flow.

| Test Case ID | Module / Function        | Test Objective                              | Input                               | White Box Technique Used | Expected Output                      | Status |
|--------------|--------------------------|---------------------------------------------|-------------------------------------|--------------------------|--------------------------------------|--------|
| WBT C-01     | User Registration Module | Verify all code paths for user registration | Valid username, email, and password | Statement Coverage       | User account is created successfully | Pass   |
| WBT C-02     | User Login Module        | Check login authentication logic            | Correct username and password       | Branch Coverage          | User is logged into the dashboard    | Pass   |
| WBT C-14     | Database Module          | Verify database connection handling         | Valid database credentials          | Statement Coverage       | Database connection established      | Pass   |
| WBT C-11     | Analytics Dashboard      | Verify no data condition                    | No student activity data available  | Branch Coverage          | No Data Available" message is shown  | Pass   |



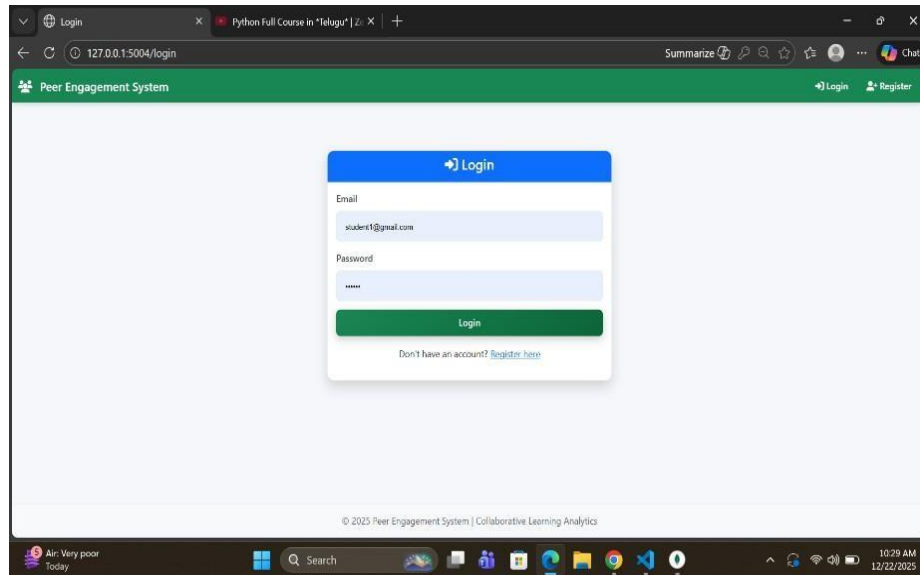
## 8 OUTPUT SCREENS



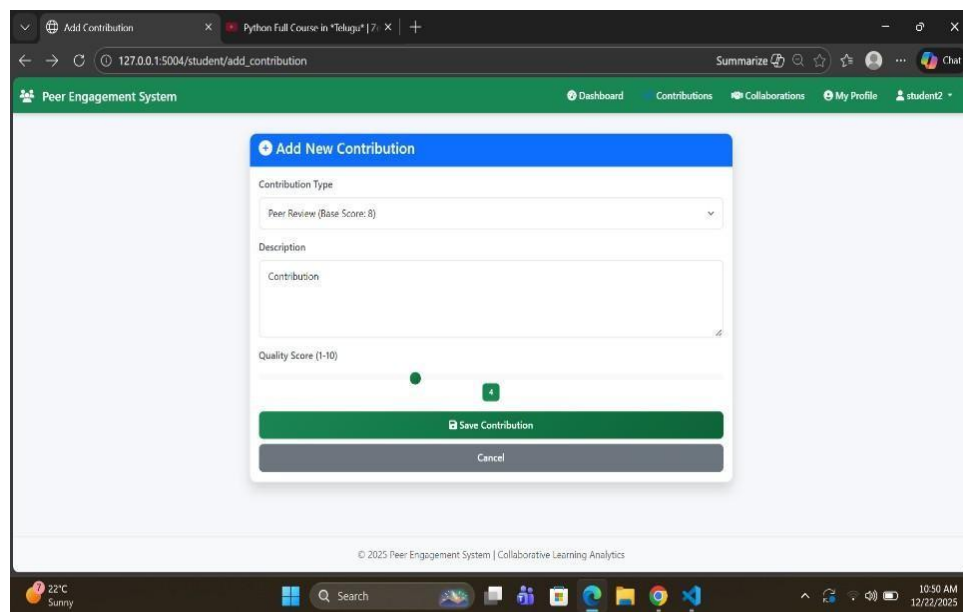
**Fig 8.1** Show the webpage for students and educators.

The screenshot shows the 'Register' page of the 'Peer Engagement System'. The header is green with the system name and 'Login' and 'Register' links. The main content area is light blue. A white registration form is centered, titled 'Register' with a person icon. The form contains the following fields: 'Username' (text input with 'student1@gmail.com'), 'Email' (text input with 'student1@gmail.com'), 'Password' (password input with '\*\*\*\*\*' and a note 'Minimum 6 characters'), 'Role' (dropdown menu with 'Student' selected), and 'Department' (text input with 'EEE'). A green 'Register' button is at the bottom of the form.

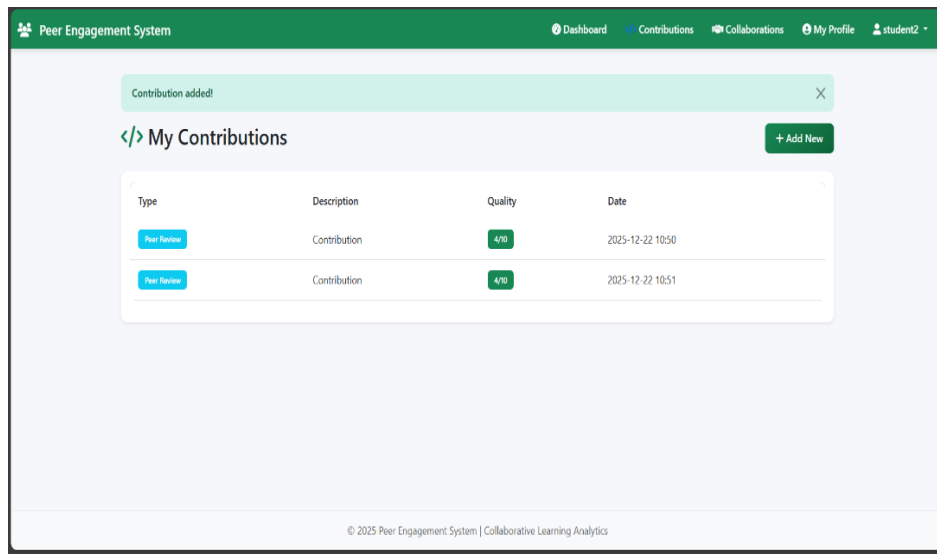
**Fig 8.2** Register page for student.



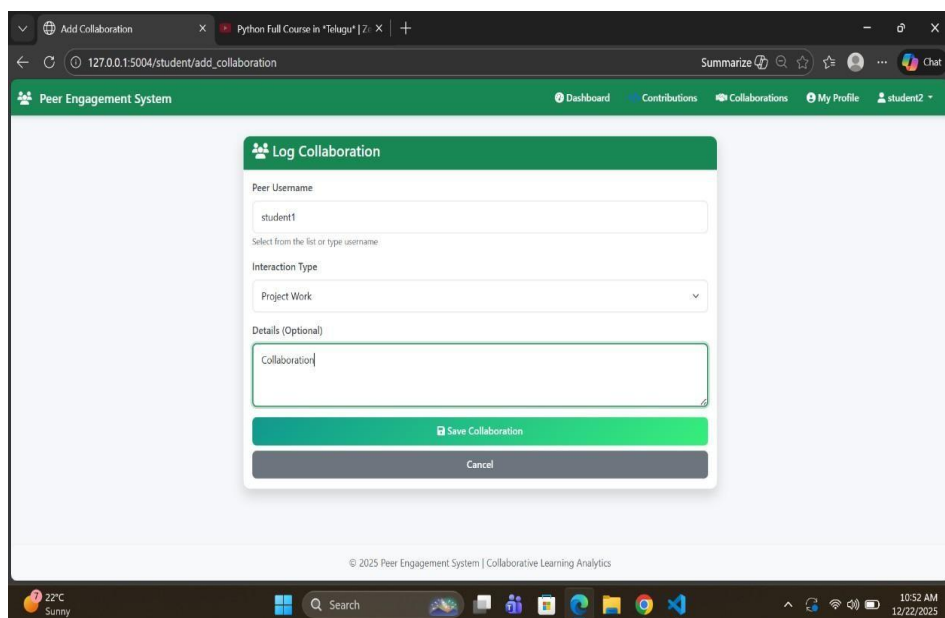
**Fig 8.3** Login page for students.



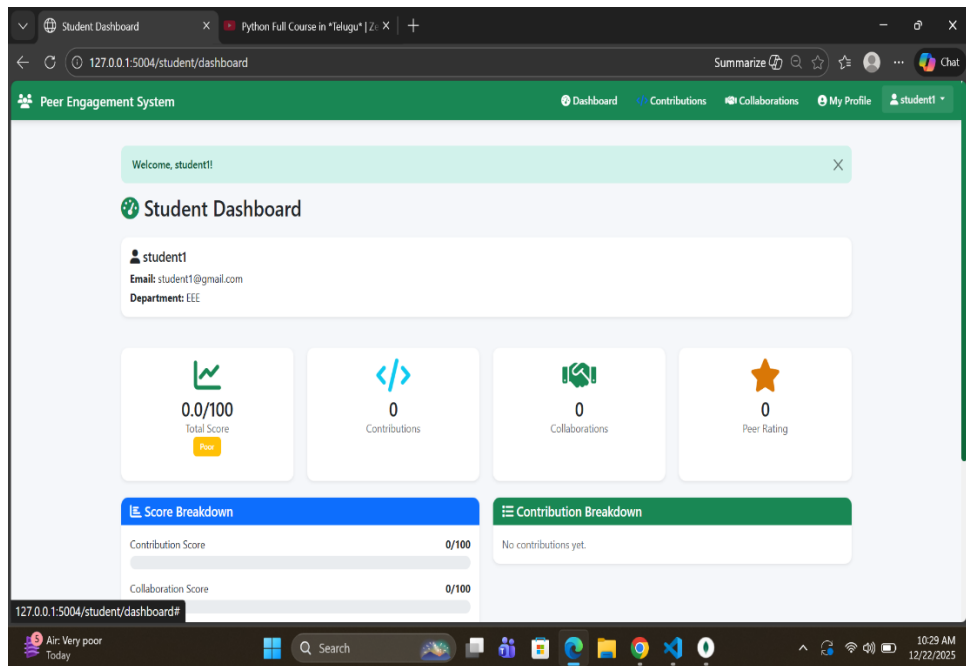
**Fig 8.4** students add contribution and save.



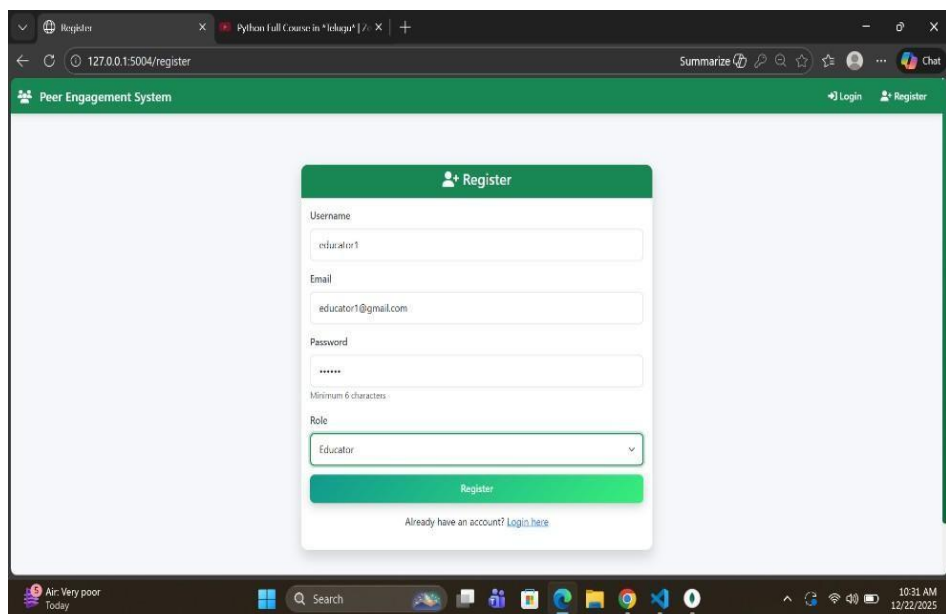
**Fig 8.5** student's contributions.



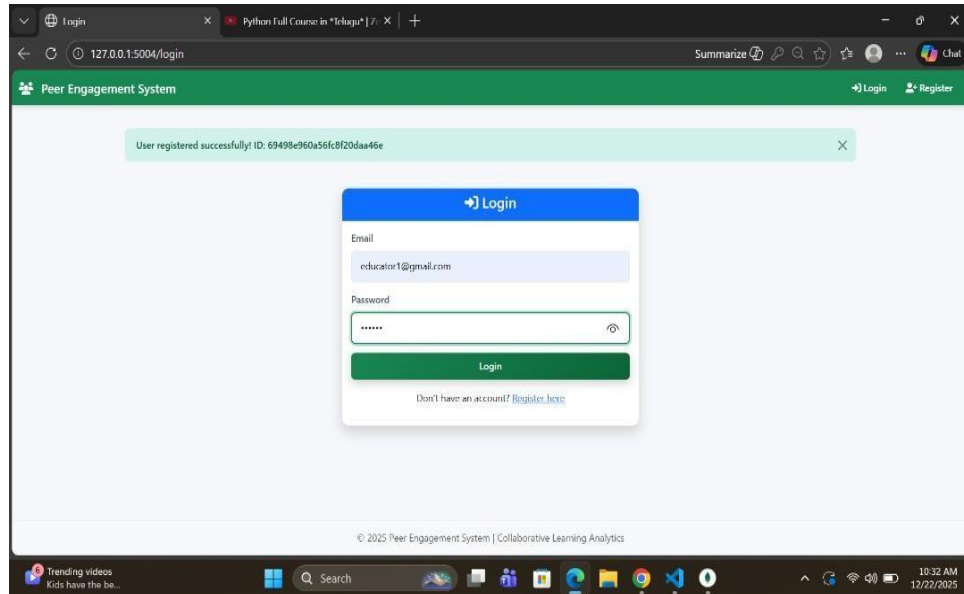
**Fig 8.6** student's collaborations.



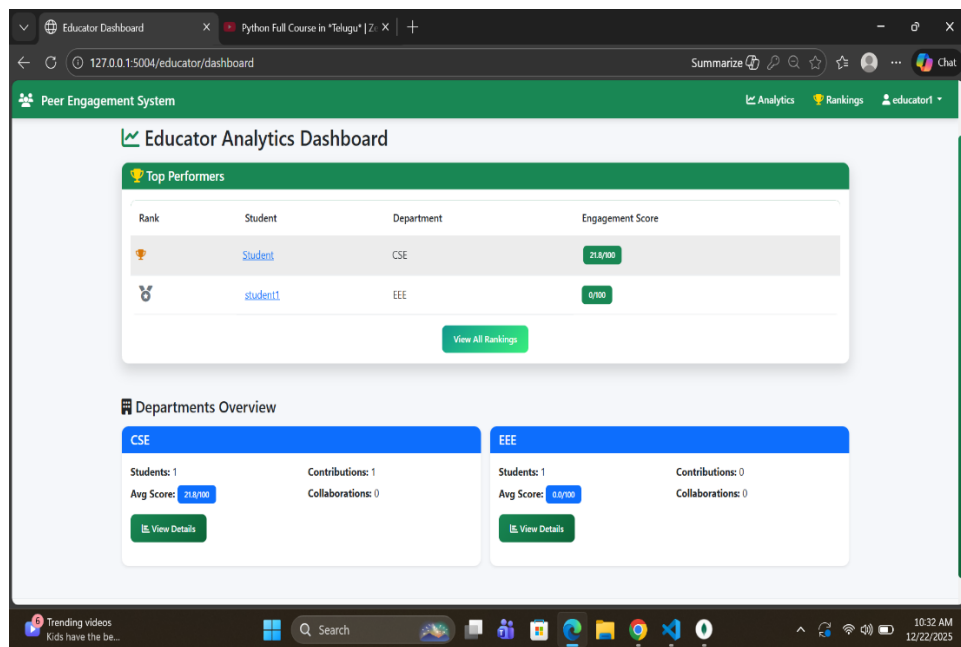
**Fig 8.7** student's Dashboard.



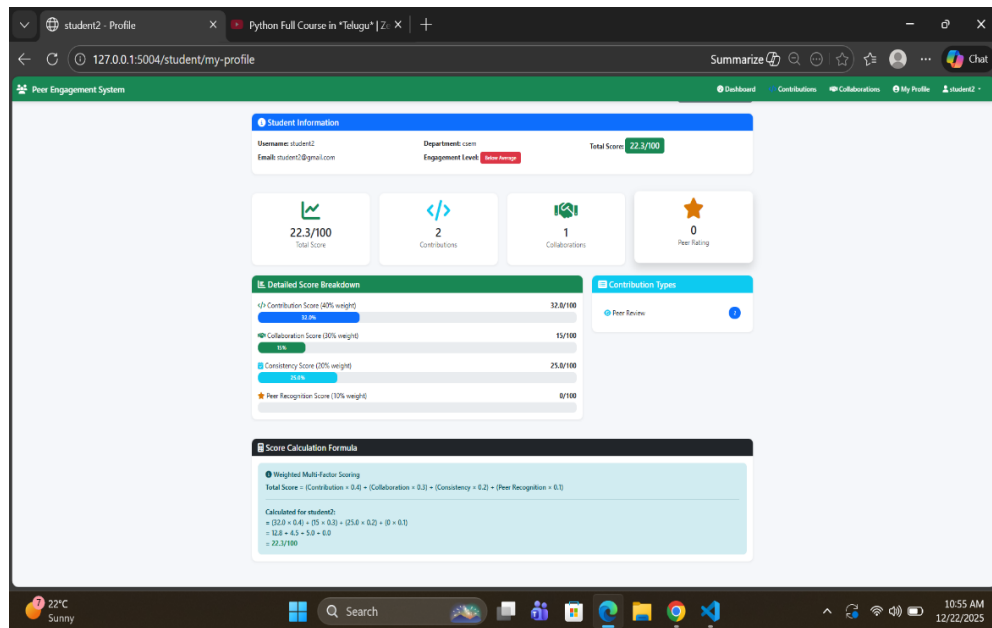
**Fig 8.8** Register page for Educators.



**Fig 8.9** Login page for Educators.



**Fig 8.10** Educators Dashboard.



**Fig 8.11** Students profile.

## 9 CONCLUSIONS

The "Peer-to-Peer Engagement Attribution and Collaborative Learning Analytics Model" highlights that while collaborative learning (CL) enhances student motivation, its effectiveness in boosting academic engagement is primarily mediated by the quality of peer support.

**Key Findings:** The model indicates that CL increases peer support, which directly boosts student engagement, particularly in smaller group settings.

**The "Pseudo-Collaboration" Trap:** A critical finding is that high interaction does not always guarantee high engagement; "Inefficient Socializers" may engage in social interaction that is detached from task-oriented goals, which can inhibit academic engagement.

**Attribution & Analytics:** The model successfully uses Latent Profile Analysis (LPA) and structural equation modeling (SEM) to attribute engagement to specific, high-quality peer interaction behaviors rather than simple participation.

**Outcome:** The model confirms that effective P2P learning requires structured, task-oriented interaction to foster academic engagement

## **10 FUTURE SCOPE**

### **10.1 Enhancements**

Enhancements for Peer-to-Peer Engagement Attribution and Collaborative Learning Analytics Model

Enhancements refer to improvements or additional features that can be added to the existing system to make it more efficient, intelligent, and useful for educators and students.

#### **10.1.1 Real-Time Analytics Dashboard**

The system can be enhanced by adding a real-time analytics dashboard that displays student participation, collaboration levels, and contribution scores instantly. This helps educators monitor student engagement during collaborative activities and identify inactive students quickly.

#### **10.1.2 AI-Based Engagement Prediction**

Artificial Intelligence and Machine Learning techniques can be integrated to predict student engagement patterns. The system can analyze previous interactions and forecast which students may become inactive or require additional support.

#### **10.1.3 Automated Feedback System**

An automated feedback module can be added to provide instant feedback to students based on their contributions, peer ratings, and collaboration performance. This encourages students to improve their participation.

#### **10.1.4 Advanced Visualization Tools**

The system can include interactive graphs, charts, and heat maps to visually represent collaboration patterns, contribution levels, and peer interactions. These visual

insights make it easier for educators to understand student behavior.

#### **10.1.5 Integration with Learning Management Systems (LMS)**

The model can be integrated with platforms such as Moodle, Google Classroom, or Canvas to collect engagement data automatically and streamline learning analytics.

### **10.2 Recommendations**

#### **10.2.1 Reference**

Frontiers in Psychology (2025). Peer collaborative learning and academic engagement.

Frontiers in Psychology (2025). Impact of collaborative learning on student engagement in college.

ScienceDirect (2022). Exploring peer effects on learner engagement in collaborative reading.

PMC (2026). Peer collaborative learning and academic engagement in....

Springer (2023). Multimodal learning analytics of collaborative patterns during pair....

Springer (2023). Learning Analytics for Peer Assessment: A Scoping Review.

[1.2.2/1.4.8] Inder science (2008). Peer-to-peer technology in collaborative learning networks.